

# Chapter 1

## Rings

### 1.1 Rings and Subrings

We now continue to generalize the familiar arithmetic operations of addition and multiplication. Just as with groups, results from the algebra of arithmetic will be extended to a host of non-numerical objects, e.g., matrices, functions, polynomials, etc.

- From here on out we will use the following notations unless stated otherwise:

+ for the “addition” operation, e.g.,  $a + b$  (instead of  $\oplus$ ).

$\cdot$  or juxtaposition for the “multiplication” operation, e.g.,  $a \cdot b$  or  $ab$  (instead of  $\otimes$ ).

“ $-a$ ” will denote the additive inverse of an element  $a$ , e.g.,  $-a + a = 0$  (instead of “ $a^{-1}$ ”).

“ $a^{-1}$ ” will denote the multiplicative inverse of an element  $a$ , e.g.,  $a^{-1}a = 1$ .

**Definition 1.** A *ring* is a nonempty set, say  $R$ , with two binary operations (usually written as multiplication and addition)<sup>1</sup> that satisfies the following:

1. If  $a, b \in R$  then  $a + b \in R$ . (closed under addition)
2.  $a + (b + c) = (a + b) + c$  (associative addition)
3.  $a + b = b + a$  (commutative addition)
4. There exists an  $0_R$  such that  $a + 0_R = 0_R + a = a$  (additive identity)
5. There exists an  $-a \in R$  such that  $a + (-a) = 0_R$  for every  $a \in R$ . (additive inverse)
6. If  $a, b \in R$  then  $ab \in R$ . (closed under multiplication)
7.  $a(bc) = (ab)c$  (associative multiplication)
8.  $a(b + c) = ab + ac$  and  $(a + b)c = ac + bc$  (distribution holds).

- Note that parts 1-5 say that a ring is an *abelian group*.<sup>2</sup>

**Example 1.** The set of integers,  $\mathbb{Z}$ , under addition and multiplication is a ring.

**Example 2.**  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$  ( $\mathbb{Z}$  modulo 4) is also a ring under addition and multiplication as in  $\mathbb{Z}_n$  for any  $n \in \mathbb{N}$ .

**Example 3.** The set of 2 by 2 matrices with real numbers as entries,

$$\mathcal{M}_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

Closure, associativity, and distribution can be shown using the usual rules of matrix multiplication and addition.

$\mathcal{M}_2(\mathbb{R})$  is commutative under addition (but not multiplication), the additive inverse is  $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$ , the additive

identity is  $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ , and the multiplicative identity is  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ .

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<sup>1</sup>But of course the operations are not necessarily addition and multiplication.  $+$  and  $\times$ ;  $\oplus$  and  $\otimes$ ; are among the different notations that may be used in varying texts.

<sup>2</sup>Commutative groups are also called abelian in honor of Norwegian mathematician Niels Abel.

**Proposition 1.** If  $a$  and  $b$  are two elements in  $R$  then  $(-a)(-b) = ab$ .

*Proof.* By definition  $a + (-a) = 0$ .  $(a + (-a))(-b) = 0$  implies  $a(-b) + (-a)(-b) = 0$  by distribution. The additive inverses of  $a(-b)$  is unique thus  $a(-b) + (-a)(-b) = 0$  implies that  $(-a)(-b) = ab$ .  $\square$

- In particular for rings like  $\mathbb{Z}$  and  $\mathbb{R}$ , the product of two negatives is a positive. Note the importance of distribution.

**Exercise 1.** Show that  $\mathbb{Z}_6$  is a ring with unity under addition and multiplication.

**Exercise 2.** Show that the set  $\sqrt{2}\mathbb{Z} = \{x : x = m + \sqrt{2}n \text{ for some } m, n \in \mathbb{Z}\}$  is a ring.

**Exercise 3.** Let  $R$  and  $S$  be rings. Define addition and multiplication on the Cartesian product  $R \times S$  by,

$$(r, s) + (r', s') = (r + r', s + s') \text{ and } (r, s)(r', s') = (rr', ss')$$

for  $r, r' \in R$  and  $s, s' \in S$ . Prove that  $R \times S$  is a ring.

**Definition 2.** A *commutative ring* is a ring  $R$  such that  $ab = ba$  for all  $a, b \in R$ .

**Definition 3.** A ring with *identity*<sup>3</sup> is a ring  $R$  with an element  $1_R$  such that  $a1_R = 1_Ra$  for all  $a \in R$ .

- It is important to check that a potential  $1_R$  is both a *right identity*,  $a1_R = a$ ; and a *left identity*,  $1_Rb = b$ .

**Example 4.**  $\mathbb{C}$ ,  $\mathbb{R}$ ,  $\mathbb{Q}$ , and  $\mathbb{Z}$  are all commutative rings with identity under the usual addition and multiplication.

**Example 5.**  $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$  where  $n \geq 2$  is a commutative ring with identity under the usual addition and multiplication (recall that  $n$  is called the *modulus*).

- To add  $x + y \pmod n$  find  $(x + y) \div n$  and find the remainder, say  $r$ , then  $(x + y) \pmod n = r$ . In the same way we can find  $(xy) \pmod n$ . For example,  $(3 \cdot 7) \pmod 8 = 5$  since  $2 \cdot 8 + 5 = 3 \cdot 7$  (divide by 8 and take the remainder).

*Proof.* Since  $r < n$ ,  $\mathbb{Z}_n$  will be closed under addition and multiplication. Parts 2, 3, 4, and 7 hold because of the normal rules of arithmetic. Which also give multiplicative commutativity. 0 and 1 are the additive and multiplicative identities respectively.

Which only leaves distribution, part 8, and additive inverse, part 5. Let  $a \in \mathbb{Z}_n$  then  $a < n$  so then  $n - a \in \mathbb{Z}_n$ , and  $a + (n - a) = 0$ . For distribution, let  $a, b, c \in \mathbb{Z}_n$ .  $a(b + c) \pmod n = (ab + ac) \pmod n = ab \pmod n + ac \pmod n$  (this was shown last semester).  $\square$

**Example 6.**  $E = \{x : x = 2n \text{ for some } n \in \mathbb{Z}\}$ , the set of even integers is a commutative ring. However it does *not* have identity.

- Keep in mind that a ring is *not* a group under multiplication. Elements in a ring do not necessarily have a multiplicative inverse, and rings do not necessarily have a multiplicative identity. Keep examples of rings in mind which do not have these properties.

**Exercise 4.** Prove that a ring that is cyclic<sup>4</sup> under addition and commutative (under multiplication).

**Theorem 1.** If a ring has an identity, it is unique.

*Proof.* Suppose to the contrary that a ring  $R$  has two identity elements 1 and  $1'$ .  $a1 = a$  for all  $a \in R$  including  $1' \in R$ . Thus  $1' \cdot 1 = 1'$  and  $1 \cdot 1' = 1$ . By the definition of a ring  $1' = 1 \cdot 1' = 1' \cdot 1 = 1$ . Therefore  $1' = 1$ , and the multiplicative identity is unique.  $\square$

**Exercise 5.** Let  $F$  be the set of all functions from  $\mathbb{R}$  to  $\mathbb{R}$ . Define addition and multiplication as

$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

Show that  $F$  is a commutative ring with identity.

**Exercise 6.** If a ring element has a multiplicative inverse, it is unique.

<sup>3</sup>The term *unity* is often used in place identity. A ring with an identity element is said to “have unity”.

<sup>4</sup>Recall that a group  $G$  is *cyclic* if for every  $x \in G$  there is some  $g \in G$  such that  $x = g + g + \dots + g$ . In which case we write  $G = \langle g \rangle$  and say that  $G$  is *generated* by  $g$ , e.g.,  $\mathbb{Z}_3 = \langle 1 \rangle = \langle 2 \rangle$  since  $2 + 2 = 1$ ,  $2 = 2$ , and  $2 + 2 + 2 = 0 \pmod 3$ .

## 1.2 Subrings

**Definition 4.** Let  $R$  be a ring and  $S$  be a subset of  $R$ .<sup>5</sup> If  $S$  is a ring under the same operations as  $R$ , then  $S$  is a *subring*.

**Example 7.**  $\mathbb{Q}$  is a subring of  $\mathbb{R}$ , and  $\mathbb{Z}$  is a subring of  $\mathbb{Q}$  and  $\mathbb{R}$ .

**Example 8.** The set of even integers is a subring of  $\mathbb{Z}$ , but the set of odd integers is not since it is not closed under addition.

The following two theorems provide shortcuts when trying to show that a subset is a subring. Particularly, The second theorem can save quite a bit of work.

**Theorem 2.** Suppose that  $R$  is a ring and  $S \subseteq R$  such that the following hold:

1. If  $a, b \in S$  then  $a + b \in S$  (closed under addition)
2. If  $a, b \in S$  then  $ab \in S$  (closed under multiplication)
3.  $0_R \in S$ .
4. If  $a \in S$  then there is an  $-a \in S$  such that  $a + (-a) = 0_R$ .

Then  $S$  is a subring of  $R$ .

*Proof.* Parts 2, 3, 7, and 8 of the Ring definition hold for all elements in  $R$  since it is a ring. So they hold for all elements in  $S$ . The remaining requirements 1-4 in the theorem satisfy the remaining requirements.  $\square$

- Rings and thus subrings must be nonempty. A requirement which is covered by 3. Since  $0_R \in S \Rightarrow S \neq \emptyset$ .

**Theorem 3.** A nonempty subset  $S$  of a ring  $R$  is a subring if whenever  $a, b \in S$ :

1.  $ab \in S$  (closed under multiplication)
2.  $a - b \in S$  (closed under subtraction)

*Proof.* If  $a - b \in S$  for all  $a, b \in S$  then  $S$  is a subgroup of  $R$  (subgroup test), and since  $R$  is additively commutative it follows that  $S$  is an abelian subgroup of  $R$  under addition (this also means that  $0_R \in S$ ).  $S$  is multiplicatively and additively associative; and  $S$  is distributive over addition because these properties hold in  $R$ .  $\square$

**Exercise 7.** Show that the set  $3\mathbb{Z} = \{x : x = 3n \text{ for some } n \in \mathbb{Z}\}$  is a subring of  $\mathbb{Z}$ .

**Exercise 8.** Show that the set  $k\mathbb{Z} = \{x : x = kn \text{ for some } n \in \mathbb{Z}\}$  for any integer  $k$  is a subring of  $\mathbb{Z}$ .

**Exercise 9.** Prove that the intersection of any two subrings of a ring  $R$  is also a subring of  $R$ .

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<sup>5</sup>Notated:  $S \subseteq R$ .